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Project Title:	Microgravity Effects on the Function of Vascular Stem Cells		
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Comments:			
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Task Description:	The negative effects of space flight on cardiovascular health has been demonstrated and documented over many years. One of the body's normal vascular repair mechanisms is the recruitment of adult stem cells from bone marrow to integrate into the vasculature. These recruited cells are involved in blood vessel wall repair and the formation of new blood vessels. This proposal seeks to study the effect of microgravity on adult stem cells. We are specifically interested in how well these cells can differentiate into vascular cells under altered environmental conditions such as microgravity. Further, we hope to study how well these cells function as healthy vascular cells once differentiated. We propose to conduct differentiation studies using a bioreactor able to expose the cells to microgravity conditions. Once differentiated, the cells will be exposed to several stimulatory environments that will demonstrate their important functional ability. Utilizing molecular biology techniques, such as gene expression studies, we hope to elucidate the effect of microgravity on these cells. It is our hope to study adult stem cells with the goal to develop countermeasures for the negative effects of space flight on vascular health. This proposal addresses one of NASA's research interest, specifically, studies of the effect of gravity and the space environment on cellular and molecular processes. In this proposal, the cellular and molecular processes under investigation are stem cell differentiation and the corresponding molecular changes the stem cells undergo.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We seek to understand the impact of microgravity on circulating stem cells, the repair cells for the vasculature. Dysfunction of these repair cells has been implicated in the onset and progression of cardiovascular disease. Our work has impacts for life on Earth and in space. We are modelling our experiments to mimic the conditions in space where there is a lack of gravity; however, we are also including control samples that are treated under normal gravity. Once an astronaut who completed time in space returns to the normal gravity of Earth, it is our goal that the work we conducted will reveal the effect of the lack of gravity on the astronaut's overall vascular health.
	Our progress has included completion of several goals: 1. Goal: Optimizing bioreactor and experimental set-up and methodology with Human Umbilical Vein Endothelial Cells (HUVECs) The experimental set-up we proposed using a commercially available rotary cell culture system (RCCS) bioreactor (Synthecon) was optimized using HUVEC cells. Our goal was to optimize various bioreactor/experimental parameters such as microcarrier bead handling and protein coating, cell seeding density and culture parameters, bioreactor rotational speed, culture maintenance and sample extraction from bioreactor chambers, as well as overall training within the experimental system. Early project challenges such as low cell adhesion on microcarrier beads and cell detachment under dynamic conditions in the rotating wall vessel (RWV) were eventually overcome by changing our experimental design. Our results show successful cell attachment and faster proliferation of HUVECs under microgravity than under static conditions. 2. Goal: Isolation and culture of human endothelial progenitor cells and differentiation into endothelial cells Once our bioreactor and experimental setup was optimized, we extended our work to the target cell population that is the focus of this work, human endothelial progenitor cells (EPC) derived from circulating blood. The human EPCs are a rare population of stem/progenitor cells that are isolated from the peripheral mononuclear cell (PBMNC) fraction of whole blood. PBMNCs were isolated from human blood obtained from donors as per Institutional Review Board (IRB) guidelines. Circulating progenitor cells contained within the PBMNC fraction of whole blood were differentiated into endothelial-like cells, as confirmed by the formation of colonies (within 2-6 weeks) with EC specific 'cobble-stone' morphology, and positive expression of EC specific proteins. 3. Goal: Study the effect of microgravity on the differentiated endothelial cells in culture We assessed the ability of these cells to adapt to microgravity conditions after short term culture and expansion. More specifically, the functionality of the differentiated ECs following exposure to simulated microgravity assessed conditions such as proliferation, release of soluble factors, changes in gene expression, and anti-thrombogenic potential was investigated. More specifically, the progress made is as follows: 3a. Cell proliferation: Initially microgravity caused a delay in proliferation of the endothelial cells. On prolonged exposure there was an increase in proliferation of microgravity-exposed cells when compared to the normal gravity cultured controls. 3b. Gene Expression of hsp70: Further we investigated the ability of HE-like cells to sense microgravity as a stressful condition and adapt to it by regulating the levels of stress proteins such as hsp70. Our results show that microgravity-cultured cells upregulated gene expression of HSPA4 after 2 days of culture. However, on prolonged exposure there was no difference in the level of HSPA4 between simulated microgravity and normal gravity. 3c. Endothelial Cell secretory functions: Secretion of Nitric Oxide (NO), Interleukin-6, and Tissue Plasminogen Activator-- Our results indicate that HE-like cells cultured under microgravity for 2 days upregulated release of amount of soluble NO when compared to the respective normal gravity controls while on prolonged exposure the amount of NO released by both microgravity cultured as well as normal gravity cultured cells were identical. Assessment of the release of soluble IL-6, a major pro-inflammatory cytokine, show that culture under microgravity conditions for 2 and 4 days led to significant decrease in secretion of soluble IL-6 in the cell culture medium when compared to normal gravity cultured controls. Quantification of the release of soluble tissue plasminogen activator (tPA) into the surrounding cell culture medium showed that tPA released by HE-like cells cultured under microgravity conditions for 2 and 4 days were significantly lower than their respective normal gravity cultured controls. 3d. Plasma clotting kinetics: To further study and confirm the direct impact of microgravity on the ability of these cells to regulate clotting, we performed a modified version of plasma clotting assay. This assay provides a robust and direct method to assess and compare the anti-thrombogenic potential of HE-like cells cultured under different gravity conditions. Once clotting was activated, progress to full clot formation was assessed spectrophotometrically by continuous monitoring of the change in absorbance over 30 min. Our results show that the kinetic profile of plasma clotting when incubated with HE-like cells cultured under simulated microgravity showed an earlier onset to clot formation than normal gravity controls. The quantification of the time taken to complete clot formation showed

Task Progress:	recalcified plasma clots faster when incubated with HE-like cells cultured under simulated microgravity as compared to normal gravity cultured HE-like cells. In summary, the results of the present study demonstrate that the populations of human blood EPC derived endothelial cells when exposed to microgravity proliferate faster, release higher amount of soluble NO, and lower amount of pro-inflammatory cytokines such as IL-6, when compared to their normal gravity cultured controls. This can be rationalized and understood as the ability of the cells to sense microgravity and adapt by upregulating gene expression of hsp70. However, these secretory as well as proliferative functions did not affect the anti-thrombogenic functions of these cells. Their anti-coagulative and anti-thrombogenic ability as assessed by both upregulation of tPA as well their ability to delay plasma clotting, were impaired on exposure to microgravity. 4. Goal: Expand to include porcine EPCs to further study the effect of microgravity on differentiation of the blood derived EPCs Due to the initial challenges we had faced with differentiating as well exposing human blood derived progenitor cells during the pre-differentation stage under simulated microgravity, our goal was to extend our work to an alternative cell source, porcine EPCs. Our lab has previously obtained porcine PBMNCs (under University of Florida IACUC approval) and differentiated these into cells with endothelial cell phenotype (PE-like cells). Circulating progenitor cells contained within the PBMNC fraction of porcine whole blood were exposed to simulated microgravity, using a RWV bioreactor. After exposing the circulating progenitor cells to 6, 12, and 18h of simulated microgravity, the cells were then cultured on fibronectin coated TCP under static conditions and their ability to differentiate into endothelial cells was assessed by the formation of colonies with EC specific ‘cobble-stone’ morphology and confirmed by immunohistochemical staining for CD31 as well as eNOS. Our results showed that longer exposure to simulated microgravity delayed the onset of differentiation of EPCs to PE-like cells. Further, we investigated the effect of this altered condition on the downstream functional abilities of the differentiated endothelial-like cells that were pre-exposed to microgravity. 4a. Colony formation: Assessed by quantifying the number of endothelial-cell colonies formed showing the onset of differentiation of EPCs to ECs. Our results show that the average rate of endothelial cell colony formation decreased with increased time of exposure to simulated microgravity. Cells exposed to microgravity for longer than 18h showed significantly reduced or no colony formation 4b. Cell Proliferation and Migration: The differentiated cells were assessed for their proliferation and migratory ability. Migration was assessed by observing and quantifying the migration of cells into an area cleared of cells. The quantification of percentage area of the scratch covered by the migrating cells indicated a direct decrease in migration ability of the EC with increased time of exposure of the progenitor cells under simulated microgravity. The effect of pre-exposure to µg on the proliferative capacity of the EC-like cells was also assessed by quantifying the change in DNA over time in cultured cells. These data collectively show that both the migratory ability and proliferative ability of ECs are impaired by pre-exposure of the EPCs to microgravity. While increased exposure time to µg impaired the migration of the microgravity exposed cells, the proliferation of these cells seem to be enhanced by increased time of microgravity pre-exposure. 4c. Endothelial cell secretory functions: We further investigated the ability of these cells to secrete soluble factors such as nitric oxide (NO) as well PGE2 (prostaglandin E2) into the surrounding cell culture medium. Our data collectively shows the ability of short term microgravity exposure of EPCs to inhibit the regulation of release anti thrombogenic molecules into the surrounding medium. However with increased time of microgravity exposure, a potential recovery effect is also observed where the 12h and 18h-µg exposed cells secrete NO and PGE2 in levels comparable to the cells cultured under normal gravity. 4d. Whole Blood and Plasma Clotting kinetics: To understand the ability of the cells to regulate clotting mechanisms, whole blood clotting and recalcified plasma clotting studies as described above were subsequently performed on these cells. Plasma recalcification clotting profiles indicated that as expected the TCP surface induced highest rate of plasma clotting within the first 90 seconds followed by 6h µg exposed PE-like cells. Following the clotting rate for 6h-µg exposed cells are the clotting times for the 12h and 18h-µg exposed cells. Normal gravity cultured cells clotted the next while the last to clot was the plasma exposed to the EC control cells. Similarly, the whole blood clotting kinetics showed that the kinetic profile of whole clotting when incubated with PE-like cells differentiated culture under simulated microgravity, showed faster clot formation than normal gravity controls. However, longer exposure times showed greater inhibition of clotting. Collectively, the data shows that within the 3 time points of µg exposure, the longer exposure times showed greater inhibition of clotting, which may be attributed to the increased ability to release soluble anti-thrombogenic factors, NO and PGE2. While both the migration and proliferation of cells were inhibited by microgravity exposure, prolonged exposure of EPCs to microgravity resulted in decrease in the migration abilities of EPC derived EC and the proliferative capacity of the ECs seemed to recover on longer time of exposure. The initial decrease and subsequent increase in release of soluble factors such as NO and PGE2 reaffirmed the possibility of recovery mechanisms being involved on prolonged exposure to weightlessness. Finally the assessment of whole blood and plasma clotting, indicated that microgravity pre exposure at the progenitor stage caused inhibition, in the ability of PE-like cells to delay blood or plasma clotting while this effect was again nulled with increase in time of exposure of EPCs to microgravity. This data collectively provides some insight into the role of short term exposure to low shear stress environment due to simulated microgravity, on cardiovascular deconditioning during space travel.
Bibliography Type:	Description: (Last Updated: 01/31/2022)
Articles in Peer-reviewed Journals	Ramaswamy V, Dirr E, Allen J. "The effect of simulated microgravity on differentiation of porcine blood derived vascular stem cells." Stem Cells and Development. 2016 Feb;25(4):329-36. Epub 2015 Dec 9. PubMed PMID: 26649474 ; http://dx.doi.org/10.1089/scd.2015.0297 , Feb-2016
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